

Machine Learning CO₂ Adsorption Prediction

Executive Summary

This report presents a partial reproduction study of a published machine learning framework for CO₂ adsorption prediction in MOFs. The original paper used deep neural networks on 342,489 MOFs, while this project used 449 MOFs with geometric and RAC descriptors.

Original Paper

The paper's best model combined Geometric, Chemical Motif Density (CMD), and AP-RDF descriptors and achieved test $R^2 = 0.958$. The dataset contained 342,489 MOFs and used a 70:10:20 train/development/test split.

Project Dataset

Geometric descriptors, RAC descriptors, and CO₂ uptake values were merged. After cleaning and matching frameworks, 449 valid MOFs remained.

Pipeline

Load descriptors → Merge datasets → Extract CO₂ uptake → Create training data → Split data → Scale features → Train DNN → Train XGBoost → Evaluate metrics.

DNN Architecture

Dense(1000)-Dropout(0.2)-Dense(1000)-Dropout(0.2)-Dense(700)-Output(1), Adam optimizer, learning rate 7.5e-4.

Results

DNN: $R^2=0.5665$, MAE=0.7230, RMSE=1.1895. XGBoost: $R^2=0.8491$, MAE=0.5136, RMSE=0.7018.

Feature Importance

Top features included lc-l-1-all, D_mc-Z-3-all, mc-S-0-all, mc-chi-3-all, ASA_m²/cm³, and ASA_m²/g.

Comparison

The original paper achieved $R^2=0.958$ using richer descriptors and a much larger dataset. This project achieved $R^2=0.8491$ using XGBoost on 449 MOFs.

Challenges

Limited dataset size, unavailable AP-RDF descriptors, inability to exactly replicate the original descriptor pipeline and training setup.

Conclusion

The work is best described as a partial reproduction study. Results are scientifically reasonable but do not numerically match the original benchmark due to differences in data scale and descriptors.